

Joins, GroupBy & Pivot

Split the data. Apply a function. Combine the results. Master this idiom and you can answer any "per-X, per-Y" question ever asked by a stakeholder.

One mnemonic for axis: "axis is the axis you COLLAPSE." Remember this for the next 55 minutes.

WHAT'S INSIDE

- 01** Split · Apply · Combine
- 02** Aggregate — one row per group
- 03** Transform — same shape as input
- 04** The four join types
- 05** Pivot and melt — long ↔ wide

WHY THIS LESSON MATTERS

Nine out of ten business questions reduce to one pattern.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Explain split-apply-combine with a concrete example.

2

Perform groupby aggregation, transform, and filter.

3

Merge two DataFrames with the four join types.

4

Pivot long data to wide, and melt back.

5

Debug axis=0 vs axis=1 with one mnemonic.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Split · Apply · Combine

The three-move dance of every
groupby

02

Aggregate — one row per...

sum, mean, count, first, and friends

03

Transform — same shape as...

Broadcast the group stat back

04

The four join types

Default is INNER. Always sanity-check
row counts before and after.

01

PART 1 OF 4

Split · Apply · Combine

The three-move dance of every groupby

CONCEPT 1 OF 4

Split · Apply · Combine

SPLIT the DataFrame into groups by one or more columns.
APPLY a function (agg, transform, filter) to each group.
COMBINE the results back into a new DataFrame or Series.
90% of analytics reduces to this pattern.

MEMORISE

S · A · C

Map every groupby you write to one of the three moves.

DEFINITION

Split · Apply · Combine

The three-move dance of every groupby

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Aggregate — one row per group

```
df.groupby("city")["sales"].sum()
df.groupby("city")["sales"].agg(["sum", "mean", "count"])
df.groupby(["city", "cat"]).agg(total=("sales", "sum"), avg=("sales", "mean"))
```

named-aggregation syntax is both clean and fast.

RETURNS

1 row per group

Perfect for reports and dashboards.

KEY POINTS

- `df.groupby("city")["sales"].sum()`
- `df.groupby("city")["sales"].agg(["sum", "mean", "count"])`
- `df.groupby(["city", "cat"]).agg(total=("sales", "sum"), avg=("sales", "mean"))`
- named-aggregation syntax is both clean and fast

CONCEPT 3 OF 4

Transform — same shape as input

```
df["city_avg"] = df.groupby("city")["sales"].transform("mean")
```

Every original row now has its group's stat.

Enables per-row comparisons — e.g. delta vs city average.

Fast and vectorised.

RETURNS

Same shape

Compute per-row deltas from group stats.

DEFINITION

Transform — same shape as input

Broadcast the group stat back

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

The four join types

Default is INNER. Always sanity-check row counts before and after.

KEY POINTS

- Default is INNER. Always sanity-check row counts before and after
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Answer a business question in one chain

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

- 1
- 2 Write the single-chain pandas expression. Compare with your partner.
- 3 Best expression goes on the board.
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which of agg / transform / filter do you use least — and why?

Q2

Where did a silent inner-join cost you in the past?

Q3

One business question you will answer with groupby this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Every groupby decomposes into:

A

Filter-Sort-Reduce

B

Map-Reduce

C

Split-Apply-Combine

D

Scan-Hash-Join

Every groupby decomposes into:



CORRECT ANSWER

Split-Apply-Combine

WHY

Split-Apply-Combine is the canonical idiom

.transform("mean") returns shape:

A (1, n)

B (groups, n)

C Same as input

D (1, 1)

.transform("mean") returns shape:



CORRECT ANSWER

Same as input

WHY

Transform broadcasts the group statistic back to the original shape

Default merge how=:

A left

B right

C inner

D outer

Default merge how=:



CORRECT ANSWER

inner

WHY

pandas defaults to INNER — rows must exist in both

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

SPLIT the DataFrame into groups by one or more columns.
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90% of analytics reduces to this pattern.

MEMORISE

S · A · C

Map every groupby you write to one of the three moves.

THE TAKEAWAY

"Clean data has no value until it is reshaped into an answer. GroupBy is the reshapener."

WHAT TO NOTICE

1

Split · Apply · Combine

2

Aggregate — one row per...

3

Transform — same shape as...

4

The four join types

Mini assignment — three questions, three chains

On any dataset you own, answer three business questions using groupby + merge + pivot.

DELIVERABLES

- Write each answer as a single chained pandas expression.
- Include a sanity-check assertion on row counts.
- Push notebook to GitHub.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Notebook URL with three clearly-labelled questions and chains.

ONE THING TO REMEMBER

“

"Clean data has no value until it is reshaped into an answer. GroupBy is the reshapener."

— AI Learning Hub

END OF LESSON 2

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 6 — Reading from CSV, Excel, SQL, JSON. The interfaces to real-world...